A FIELD EXTENSION IS A FILTERED COLIMIT OF GLOBAL COMPLETE INTERSECTIONS, AND OF SMOOTH ALGEBRAS WHEN SEPARABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (preamble.tex, 217 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted proof refers to a same-chapter statement by label; it is recorded as a navigation stub pointed at its \label line, and the argument consuming it stays inside the counted span.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- Filtered colimits over the finite subsets of an algebra. Used at the start of the counted proof (lines 44752-44754). Reduces the statement to finding one subalgebra containing a given finite set with the required property.
- Separable extensions generated by one element. Invoked in the counted proof at lines 44755-44760. Supplies the smooth algebra in the separable case.

Source reference. Original lemma 037X: sources/stacks-algebra/pinned-original-source.tex, source lines 44729–44729. Navigation pointer for \ref{lemma-localization-smooth-separable} used inside the counted span (line 44756); the argument that consumes it is retained in the span.

Lemma 0.1. *Let K/k be a field extension. Then K is a filtered colimit of global complete intersection algebras over k. If K/k is separable, then K is a filtered colimit of smooth algebras over k.*

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Suppose that $E \subset K$ is a finite subset. It suffices to show that there exists a k subalgebra $A \subset K$ which contains E and which is a global complete intersection (resp. smooth) over k. The separable/smooth case follows from Lemma 037X. In general let $L \subset K$ be the subfield generated by E. Pick a transcendence basis $x_1, \ldots, x_d \in L$ over k. The extension $L/k(x_1, \ldots, x_d)$ is finite. Say $L = k(x_1, \ldots, x_d)[y_1, \ldots, y_r]$. Pick inductively polynomials $P_i \in k(x_1, \ldots, x_d)[Y_1, \ldots, Y_r]$ such that $P_i = P_i(Y_1, \ldots, Y_i)$ is monic in Y_i over $k(x_1, \ldots, x_d)[Y_1, \ldots, Y_{i-1}]$ and maps to the minimum polynomial of y_i in $k(x_1, \ldots, x_d)[y_1, \ldots, y_{i-1}][Y_i]$. Then it is clear that $P_1, \ldots, P_r$ is a regular sequence in $k(x_1, \ldots, x_r)[Y_1, \ldots, Y_r]$ and that $L = k(x_1, \ldots, x_r)[Y_1, \ldots, Y_r]/(P_1, \ldots, P_r)$. If $h \in k[x_1, \ldots, x_d]$ is a polynomial such that $P_i \in k[x_1, \ldots, x_d, 1/h, Y_1, \ldots, Y_r]$, then we see that $P_1, \ldots, P_r$ is a regular sequence in $k[x_1, \ldots, x_d, 1/h, Y_1, \ldots, Y_r]$ and $A = k[x_1, \ldots, x_d, 1/h, Y_1, \ldots, Y_r]/(P_1, \ldots, P_r)$ is a global complete intersection. After adjusting our choice of h we may assume $E \subset A$ and we win. $\square$